ALZpath Simoa pTau-217 v2 assay on Insight46 phase 1 and 2

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Document details

Categories of variables	Blood, urine and other samples
Summary of work undertaken	**Category:** Derivation **Description:** Overview The returned data are measures of a blood-based biomarker for Alzheimer's disease neuropathology called p-tau 217. Plasma samples collected at 69-year home visits were assayed using the ALZpath Simoa p-tau 217 assay version 2 at the UK DRI Biomarker Factory at UCL in spring 2024. 952 samples were assayed: this subset of approximately half of all available 69-year home visit plasma samples was selected to prioritise assaying of samples from participants who had not enrolled in Insight 46 phases 1 and 2 (in which plasma p-tau 217 had already been measured). In samples from those who had not attended Insight 46 phases 1 and 2, there was also preferential selection for individuals who had died following the 69-year home visits, in order to ascertain p-tau 217 concentrations for individuals who would have no prospect of joining Insight 46 during phase 3 (where p-tau 217 assaying is planned).
Names of source variables	Conc
Output variables	ptau217_pl_i46p1, ptau217_lloq_i46p1, ptau217_pl_i46p2, ptau217_lloq_i46p2, ptau217_csf_i46p2
Papers using these variables	Papers forthcoming Technical paper on ALZpath assay v2:

Overview

The returned data are measures of a blood-based biomarker for Alzheimer's disease neuropathology called p-tau 217. Plasma samples collected at 69-year home visits were assayed using the ALZpath Simoa p-tau 217 assay version 2 at the UK DRI Biomarker Factory at UCL in spring 2024. 952 samples were assayed: this subset of approximately half of all available 69-year home visit plasma samples was selected to prioritise assaying of samples from participants who had not enrolled in Insight 46 phases 1 and 2 (in which plasma p-tau 217 had already been measured). In samples from those who had not attended Insight 46 phases 1 and 2, there was also preferential selection for individuals who had died following the 69-year home visits, in order to ascertain p-tau 217 concentrations for individuals who would have no prospect of joining Insight 46 during phase 3 (where p-tau 217 assaying is planned).

Assay result adjustments

We provide three adjustments to the raw p-tau 217 assay results: i) reassignment of values under the lower limit of quantification (0.06 pg/ml) to 0.06pg/ml; ii) adjustment of values from (i) for assay variation across plates; iii) adjustment of values from (ii) to bridge from the testing of 69-year home visit samples to the testing of plasma p-tau 217 values from Insight 46, also undertaken with the ALZpath Simoa assay (version 1 or 2?)

As part of the assaying of p-tau 217 on 69-year home visit samples, 18 plasma samples from Insight 46 were also assayed to allow for bridging of these new data to former p-tau 217 values generated on Insight 46 samples. 2 pooled plasma control samples (also used in the previous Insight 46 assaying) were also included in the assaying. Together, these 20 plasma samples were used to calculate bridging of the 69-year home visit assay results to Insight 46 assay results. The values from the 18 individual bridging samples were all from singlicate analysis in Insight46 (adjusted) and were the means of duplicate analysis in NSHD (adjusted). The values from the pooled plasma controls were means calculated from across all the 5 runs in Insight46 and up to 16 plates in NSHD.

2 pooled CSF controls were also plated across each of the 16 plates and were used to derive the plate-specific internal adjustment factor. They had also previously been quantified in Insight 46 but they were not used for the bridging calculation because they were manually diluted with different dilutions in Insight46 vs NSHD, and although they were very similar once adjusted for dilution, the dilution-adjusted concentrations are in a different range and would unduly influence correction based on linear regression if applied.

For bridging, a univariable linear regression model was applied to derive the conversion factor as follows: y = bx where y = Insight46 value, x = NSHD value and b = bridging factor. The constant was fixed at zero in this model, as zero should hold true across both quantifications given that it is the same assay.

Assay results provide absolute quantification, with units being pg/ml.

Variables in this document

5 high-confidence matches

Variable	Label	How it was found
Sub-Studies [57] › Insight46 [699] › Biological samples [6037] › Blood biomarkers [60372]		
ptau217_csf_i46p2	Cerebrospinal fluid (CSF) ALZpath p-tau217 (pg/mL) - Insight Phase 2	topsheet field
ptau217_lloq_i46p1	Indicator for whether plasma ALZpath raw value was below LLOQ - Insight Phase 1	topsheet field
ptau217_lloq_i46p2	Indicator for whether plasma ALZpath raw value was below LLOQ - Insight Phase 2	topsheet field
ptau217_pl_i46p1	Plasma ALZpath p-tau217 (pg/mL) - Insight Phase 1	topsheet field
ptau217_pl_i46p2	Plasma ALZpath p-tau217 (pg/mL) - Insight Phase 2	topsheet field

1 medium-confidence match

Variable	Label	How it was found
Questionnaire data [55] › Health and medical history [510] › Medical conditions [5101] › Other medical conditions [51011]		
blood	Have you had anaemia in the last 10 years - at age 43 years	prose scan